

# Executive Summary: Immediate Pre-Freeze Banking Stress Across Two Anaerobic Strains

## Purpose

Evaluated whether **growth phase at banking** and **freezing-medium composition** affect immediate regrowth of anaerobic bacteria after the standard banking workflow. Two strains were tested: **Bacteroides thetaiotaomicron** and **Fusicatenibacter saccharivorans FB100286**. Cells were harvested at mid-log, late-log, or late stationary phase, exposed to freezing media for ~30 minutes at room temperature, and then assessed by OD600 growth curves across dilutions. This experiment measures the effect of the **pre-freeze handling window** after glycerol addition, not long-term frozen survival.

## Key Findings

The 30-minute exposure to freezing media is not neutral. It can measurably affect regrowth before samples are ever frozen.

Across both strains, **15% glycerol in media was the most favorable condition tested**. It generally supported faster and more consistent outgrowth than higher-glycerol conditions and avoids the added stress of placing cells into glycerol-water.

**25% glycerol appeared more stressful than 15% glycerol**, especially for *B. thetaiotaomicron*. This is consistent with the fact that increasing glycerol from 15% to 25% v/v represents a large increase in solute/osmotic burden, not a minor formulation change.

The effect of growth phase was strain-dependent but important. ***B. thetaiotaomicron* showed strong sensitivity to banking condition**, especially in log-phase cultures exposed to 25% glycerol. ***F. saccharivorans* showed a weaker cryomedium effect**, but late stationary-phase cultures regrew less robustly, suggesting that those cells may already have been entering a compromised/death-phase state before banking.

## Interpretation

The data support a model where banking performance is driven by both:

1. **Cell physiology at harvest**
2. **Cryomedium-induced stress during the pre-freeze handling window**

Log-phase cells may have strong growth potential but can be vulnerable to sudden glycerol/osmotic stress. Late stationary cells may be more stress-adapted in some cases, but if held too long, they may already have reduced regrowth capacity. This points to **early stationary phase** as the likely optimal harvest window.

## Working Conclusion

The most promising general banking condition based on these data is:

**Early stationary-phase harvest + 15% glycerol prepared in growth media**

This condition should reduce pre-freeze cryomedium shock while avoiding the reduced fitness seen in late stationary/death-phase cultures.

## Next Step

The current conclusions should be tested against the matched samples stored for **1 week at -80°C**. If the same trends hold after freezing, the next focused experiment should compare **early vs late stationary phase**, using **15% glycerol in media** as the primary cryomedium.